THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 9 – Greek Letters (Delta Hedging)

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Lecture Outline

- **The Greek letters**
- Introduction to Delta
- Delta hedging
- Delta hedging and BSM
- Delta hedging with Forward contract
- Delta hedging with Futures contract
- Issues encountered with Delta hedging

- Reading: Chapter 17

The Greek Letters

Financial institutions that trade in options face the problem of how to manage their risk.

One way in which managers can hedge their positions is through the use of the Greek letters.

What are the Greek letters:

- Delta, Gamma, Vega, Theta and Rho.
- They are the partial derivatives of the BSM, each with respect to a different variable.

The Greek Letters

Delta: a measure of an option's sensitivity to changes in the price of the underlying asset.

Gamma: a measure of the Delta's sensitivity to changes in the price of the underlying asset.

Vega: a measure of an option's sensitivity to changes in the volatility of the underlying asset.

Theta: a measure of an option's sensitivity to time decay. Time decay refers to the gradual loss in an option's value as it gets closer to its expiration date.

Rho: a measure of an option's sensitivity to changes in the risk-free interest rate.

For this week's lecture, we focus on Delta.

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Introduction to Delta

Delta is defined as the rate of change of the option price with respect to the price of the underlying asset.

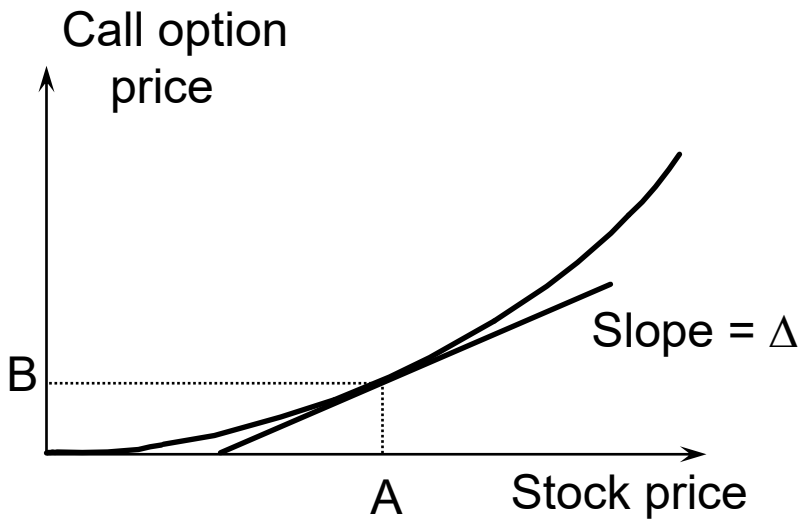
- It is the slope of the curve that relates the option price to the underlying asset.

Assume that the Delta of a call option on a stock is 0.5.

- This means that when the stock price changes by a small amount, the option price changes by approximately 50% of that amount.

Introduction to Delta

The diagram below, shows the relationship between a call price and the underlying stock price.



Introduction to Delta

When the stock price corresponds to point A, the option price corresponds to point B, and Δ is the gradient indicated.

We can see that the relationship between the stock price and the call option is not a straight line. Hence, the value of Delta will change with changes in the stock price.

In general, the Delta of a call option equals:

$$\frac{\delta c}{\delta S}$$

Where δS is a small change in the stock price and δc is the resulting change in the call price.

Introduction to Delta

Example of calculating Delta:

- Assume that a stock has a current spot price of $10, and a call option on that same stock is valued at $0.20.
- On one day, the spot price of the stock increases by 2 cents to $10.02, and the call option price increases by 0.8 cents to $0.208.
- The Delta for the option price is 0.8/2 = 0.4.
- Thus, when the stock price changes by a small amount, the option price changes by about 40% of that amount.

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Delta Hedging

A financial institution that trades in options, faces the risk that they may lose money on their options if the price of the underlying asset changes.

- Delta allows us to measure how option prices will change when the underlying asset price changes.

By using our measure of Delta, we can take a position in the underlying stock, so that the combined position in our options and the underlying stock is no longer sensitive to changes in the underlying asset price.

- Such a position is called a Delta Neutral portfolio.

Delta hedging involves maintaining a Delta neutral portfolio through time.

Delta Hedging

Example:

- A stock is currently trading at $100 per share.
- A call option at the current stock price is trading at $10 per option.
- You have sold 2,000 call option contracts. (i.e. you have taken a short position on the call options).
- The Delta of the call option on the stock is 0.6.
- If the share price rises by $1, the price of the call option will rise by $0.60.
- If you wanted to close out your position in the option, you would have to pay someone $10.60 to take your short position in the call. Therefore, with each $1 increase in the share price you will have lost $0.60 on the call option (as you are the writer).

- How do you create a Delta neutral portfolio to manage this risk?

Delta Hedging

Your current exposure is:

$$-0.6 \times 2{,}000 = -1{,}200$$

Therefore, to create a Delta neutral portfolio, you need to take an offsetting position of +1,200.

- You go long in the underlying stock.
- Each share will have a Delta of 1.

Therefore, to achieve a Delta neutral position, you will need to go long in 1,200 shares of the stock.

Delta Hedging

Examine 2 scenarios to see if your hedge works.

- The stock price goes up by $1; and,
- The stock price falls by $1.

If the stock price goes up to $101:

- The call price goes up to $10.60
- Gain on the stock position:

  $1x1,200 shares=$1,200
- Loss on the call option:

  -$0.6x2,000=-$1,200
- Your net position is zero.

Delta Hedging

If the stock price goes down to $99:

- The call price goes down to $9.40
- Loss on the stock position:
  - -$1x1,200 shares=-$1,200
- Gain on the call option:
  - $0.6x2,000=-$1,200
- Your net position is zero.

Thus you have achieved a Delta neutral portfolio, as your portfolio is no longer sensitive to stock price movements.

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Delta Hedging and BSM

The BSM can be used to calculate Delta. $N(d_1)$ can be used to calculate the Delta of European call and put options.

The Delta of a long European call option on a non-dividend paying stock, is $N(d_1)$.

- To hedge a long position in a European call on a stock that does not pay a dividend, we take a short position of $N(d_1)$ shares of stock.
- To hedge a short position in a European call on a stock that does not pay a dividend, we take a long position of $N(d_1)$ shares of stock.

Delta Hedging and BSM

The Delta of a long European put option on a non-dividend paying stock, is $N(d_1) - 1$.

- To hedge a long position in a European put on stock that does not pay a dividend, we take a long position of $1 - N(d_1)$ shares of stock.
- To hedge a short position in a European put on a stock that does not pay a dividend, we take a short position of $1 - N(d_1)$ shares of stock.
- Where d_1 is defined as:

$$d_1 = \frac{\ln(S_0/K)+(r+\sigma^2/2)T}{\sigma\sqrt{T}}$$

Delta Hedging and BSM

Call Options:

The delta of a long European Call Option on a non-dividend paying stock is $N(d_1)$.

Long in a call:

- This $N(d_1)$ is the delta of a LONG position in a call option.
- If you have a look at the N(x) tables, you will notice that $N(d_1)$ always has a positive value.
- Therefore if you are in the long position in a call option, the delta of your portfolio will be positive.
- To have a delta neutral portfolio you must combine your call option position with an asset that has a negative delta.
- We know that the delta of a stock is +1.
- Therefore, if you are long in a call option, to have a delta neutral portfolio you must be short in the underlying stock.
- Being short in the underlying stock will give you a delta which is negative.
- Thus, if you are long in a call option, to have a delta neutral portfolio you must be short $N(d_1)$ in the underlying stock.

Delta Hedging and BSM

Short in a call:

- We know that $N(d_1)$ is the delta of a long position in a call.
- If you are short in a call then the delta of your portfolio will be $-N(d_1)$
- Therefore, if you are short in a call then the delta of your portfolio must be negative.
- To have a delta neutral portfolio you must combine your call option position with an asset that has a positive delta.
- We know that the delta of a stock is +1.
- Therefore, if you are short in a call option, to have a delta neutral portfolio you must be long in the underlying stock.
- Being long in the underlying stock will give you a delta which is positive.
- Thus, if you are short in a call option, to have a delta neutral portfolio you must be long $N(d_1)$ in the underlying stock.

Delta Hedging and BSM

Put Options:

In the lecture notes we are told that the delta of a long European Put Option on a non-dividend paying stock is $N(d_1) - 1$.

Long in a put:

- This $N(d_1) - 1$ is the delta of a LONG position in a put option.
- If you have a look at the N(x) table, you will notice that $N(d_1) - 1$ will always have a negative value.
- Therefore, if you are in the long position in a put option, the delta of your portfolio will be negative.
- To have a delta neutral portfolio you must combine your put option position with an asset that has a positive delta.
- We know that the delta of a stock is +1.
- Therefore, if you are long in a put option, to have a delta neutral portfolio you must be long in the underlying stock.
- Being long in the underlying stock will give you a delta which is positive.
- Thus, if you are long in a put option, to have a delta neutral portfolio you must be long $1-N(d_1)$ of the stock.

Delta Hedging and BSM

Short in a put:

- We know that $N(d_1) - 1$ is the delta of a LONG position in a put option.
- If you are short in a put, then the delta of your portfolio must be

$-(N(d_1)-1)$

- Therefore, if you are short in a put, the delta of your portfolio must be positive.
- To have a delta neutral portfolio you must combine your put option position with an asset that has a negative delta.
- We know that the delta of a stock is +1.
- Therefore, if you are short in a put option, to have a delta neutral portfolio you must be short in the underlying stock.
- Being short in the underlying stock will give you a delta which is negative.
- Thus, if you are short in a put option, to have a delta neutral portfolio you must be short $1-N(d_1)$ of the stock.

Delta Hedging and BSM

Example:

- Given the following information, how would you implement a Delta hedge?
  - You sold call options for the purchase of 50,000 shares of stock with a strike price of $30;
  - The call options have 6 months to expiry;
  - The current share price is $29;
  - The risk-free rate is 5%; and,
  - Volatility is estimated at 25%.

Delta Hedging and BSM

You currently hold a short position in the call options.

To achieve a Delta neutral portfolio, you must take a long position of $N(d_1)$ shares in the underlying stock.

$$d_1 = \frac{\ln(\frac{29}{30}) + (0.05 + \frac{0.25^2}{2}) \times (\frac{6}{12})}{0.25\sqrt{\frac{6}{12}}} = 0.04$$

$$N(d_1) = 0.5160$$

Therefore, you should take a long position of:

0.5160 x 50,000 = 25,800 shares of the underlying stock.

Delta Hedging and BSM

The Delta of a long European call option on an asset paying a dividend yield at rate q, is $e^{-qT}N(d_1)$.

Where d_1 is defined as:

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

To hedge a short position in a European call on an asset that pays a dividend at rate q, we take a long position of $e^{-qT}N(d_1)$ of that asset.

To hedge a long position in a European call on an asset that pays a dividend at rate q, we take a short position of $e^{-qT}N(d_1)$ of that asset.

Delta Hedging and BSM

The Delta of a long European put options on an asset paying a dividend yield at rate q, is $e^{-qT}[N(d_1) - 1]$.

- To hedge a short position in a European put on an asset that pays a dividend at rate q, we take a short position of $e^{-qT}[1-N(d_1)]$ of that asset.
- To hedge a long position in a European put on an asset that pays a dividend at rate q, we take a long position of $e^{-qT}[1-N(d_1)]$ of that asset.

Delta Hedging and BSM

When the asset is a stock index, these formulae are correct with q equal to the dividend yield on the index (i.e. $e^{-qT}\Delta$).

When the asset is a currency, they are correct with q equal to the foreign risk-free rate r_f (i.e. $e^{-rfT}\Delta$).

When the asset is a futures contract (i.e. a European futures option), they are correct with q equal to the risk-free rate r (i.e. $e^{-rT}\Delta$).

Delta Hedging and BSM

Example:

- How would you implement a Delta hedge with the following information:
  - An Australian bank has sold 6 month put options on USD 1 million with a strike price of 1.25 (USD1 = AUD1.25);
  - The current exchange rate is 1.27;
  - The risk-free interest rate in Australia is 5% p.a.;
  - The risk-free interest rate in the USA is 2% p.a.; and,
  - The volatility of the USD is 20%.

Delta Hedging and BSM

The bank currently holds a short position in the put options.

To achieve a Delta neutral portfolio, the bank must take a short position of $e^{-rf\,T}[1-N(d_1)]$ in the currency.

$$d_1 = \frac{\ln(S_0/K)+(r-r_f+\sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_1 = \frac{\ln(1.27/1.25)+(.05-0.02+0.2^2/2)\times\frac{6}{12}}{0.2\sqrt{\frac{6}{12}}} = 0.29$$

$$N(d_1) = 0.6141$$

$$\Delta = (1-0.6141)\times e^{-.02\times\frac{6}{12}}$$
$$= 0.382$$

To make the position Delta neutral, the bank must therefore add a short USD position of USD1 million x 0.382.

Therefore, the bank goes short USD 382,000.

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Delta Hedging with Forward Contracts

Delta can be applied to other financial instruments than options.

We know that the payoff for a forward contract is:

$$S_0 - Ke^{-rT}$$
$$=(F-K)e^{-rT}, where\ Fe^{-rT} = S_0$$

- K is the delivery price and T is the forward contract's time to maturity.

Delta Hedging with Forward Contracts

When the price of the stock changes by δS, with all else remaining the same, the value of the forward contract on the stock also changes by δS.

The Delta of a long forward contract on one share is thus always 1.

A short forward contract on one share can be hedged by going long on one share (buying a share).

A long forward contract on one share can be hedged by shorting one share (selling a share).

Delta Hedging with Forward Contracts

The value of a forward contract with a dividend yield of q is

$$f = S_0 e^{-qT} - K e^{-rT}$$

A forward contract on an asset paying a dividend yield at rate q, has a Delta of:

$$e^{-qT}$$

For a stock index, q should be set equal to the dividend yield on the index.

For a currency, q should be set equal to the foreign risk-free interest rate.

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Delta Hedging with Futures Contracts

When the price of the stock changes by δS, with all else remaining the same, the futures price changes by $\delta S e^{rT}$.

Since futures contracts are settled daily, the holder of a long futures position makes an immediate gain of this amount. The delta of a futures contract is therefore e^{rT}.

Delta Hedging with Futures Contracts

The underlying asset is a non-dividend-paying stock:

$$\Delta = e^{rT}$$

The underlying asset pays a dividend yield q:

$$\Delta = e^{(r-q)T}$$

The underlying is a currency with the foreign risk-free rate, r_f:

$$\Delta = e^{(r-r_f)T}$$

Delta Hedging with Futures Contracts

You have been employed by an Australian option dealer to keep an eye on its exposure to currency options. One area of concern that you have identified is the dealer's exposure to pound sterling.

Currently, the dealer is ensuring that its portfolio of 11-month put options on pound sterling is kept delta neutral by making a short position in 978,000 pounds sterling.

Rather than using a position in pound sterling, you are advocating that bank takes a position in 11-month futures contracts on pound sterling.

Further, you have identified that the risk-free rates of interest in Australia are 6% p.a., and 7.5% p.a. in the U.K. (both continuously compounded).

If the dealer were to take your advice, how could 11-month futures contracts on pound sterling be used to make the dealer's position in currency options delta neutral?

Delta Hedging with Futures Contracts

We know that the delta of futures contract is:

$$\Delta = e^{rT}$$

Given that this is the futures contract written on foreign currency, the delta is:

$$\Delta = e^{(r-r_f)T}$$

We were told that the domestic (i.e., Australian) risk-free rate is 6% p.a., and the foreign (i.e., UK) risk-free rate is 7.5% p.a.

$$\Delta = e^{(0.06-0.075)\times\frac{11}{12}} = 0.9863441$$

Given that our exposure is 978,000 pounds sterling:

$$978{,}000/0.9863441 = 991{,}540$$

Hence, we need to short sufficient futures contracts written on pound sterling to cover 991,540 pounds.

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Issues Encountered in Delta Hedging

As mentioned in the lecture, Delta is not constant.

With large movements in the share price, the value of Delta will change.

To maintain a Delta neutral portfolio, traders must adjust their position to take into account changes in Delta.

This process can be time consuming as the portfolio must be rebalanced every time Delta changes.

The process can be quite expensive. If a trader is short in call options:

- When the spot price rises, they must buy more shares to maintain a Delta neutral portfolio.
- When the spot price falls they must sell shares to maintain a Delta neutral portfolio.
- This strategy is costly as the trader must buy high, and sell low.

Next Week

We are going to look at another type of derivatives – **Swaps.**